

Comprehensive Exam Style Questions Midterm 1

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1 Spring 2025

Problem 1. Let $n \geq 3$ and let $i \in \{1, 2, \dots, n\}$.

- (a) Show that $H = \{\sigma \in S_n : \sigma(i) = i\}$ is a subgroup of S_n .
- (b) In S_8 , let $\alpha = (123)(45)(67)$. Determine the number of elements in $C_{S_8}(\alpha)$, the centralizer of α in S_8 .

Problem 2. (a) Let G be a group and let $x \in G$ with $|x| = n$. Prove

$$x^k = e \iff n \mid k.$$

- (b) Prove that, in a finite Abelian group, for any $x, y \in G$, we have

$$|xy| \mid \text{lcm}(|x|, |y|).$$

2 Fall 2024

Problem 3. Let G and G' be groups and let $\varphi : G \rightarrow G'$ be a surjective group homomorphism. Prove that if T is a subgroup of G' , then $\varphi^{-1}(T)$ is a subgroup of G .

3 Spring 2024

Problem 4. Let G be a group and let $a \in G$ such that $|a|$ is finite.

- (a) Let $k \in \mathbb{Z}$. Prove that $a^k = e$ if and only if $|a|$ divides k .
- (b) Let G_1 and G_2 be finite groups and let $(a, b) \in G_1 \times G_2$. Prove $|(a, b)| = \text{lcm}(|a|, |b|)$.

4 Fall 2023

Problem 5. Let S_n be the symmetric group of order n .

- (a) Find the smallest n for which S_n contains an element of order 12. Justify your answer.
- (b) Does S_8 contain an element of order 14? Justify your answer?
- (c) Find an element τ in S_8 of maximal order. Justify your answer.

- (d) How many conjugates does $\sigma = (12)(34)(56)$ have in S_8 ?
- (e) Let σ be as in part (d). Determine the order of the centralizer $C_{S_8}(\sigma)$ in S_8 .

5 Fall 2022

Problem 6. Let G be a group.

- (a) For $x, y \in G$, we define $x \sim y$ if and only if $x = y$ or $x = y^{-1}$. Prove $\sim$ is an equivalence relation on G .
- (b) If G has even order, prove that the number of elements of G of order 2 is odd.

6 Fall 2021

Problem 7. Consider the permutation $\alpha = (12345)(678)$ in S_8 .

- (a) Prove that $C_{S_8}(\alpha) = \langle \alpha \rangle$.
- (b) Find a permutation $\sigma \in S_8$ such that $\sigma\alpha\sigma^{-1} = \alpha^{-1}$.